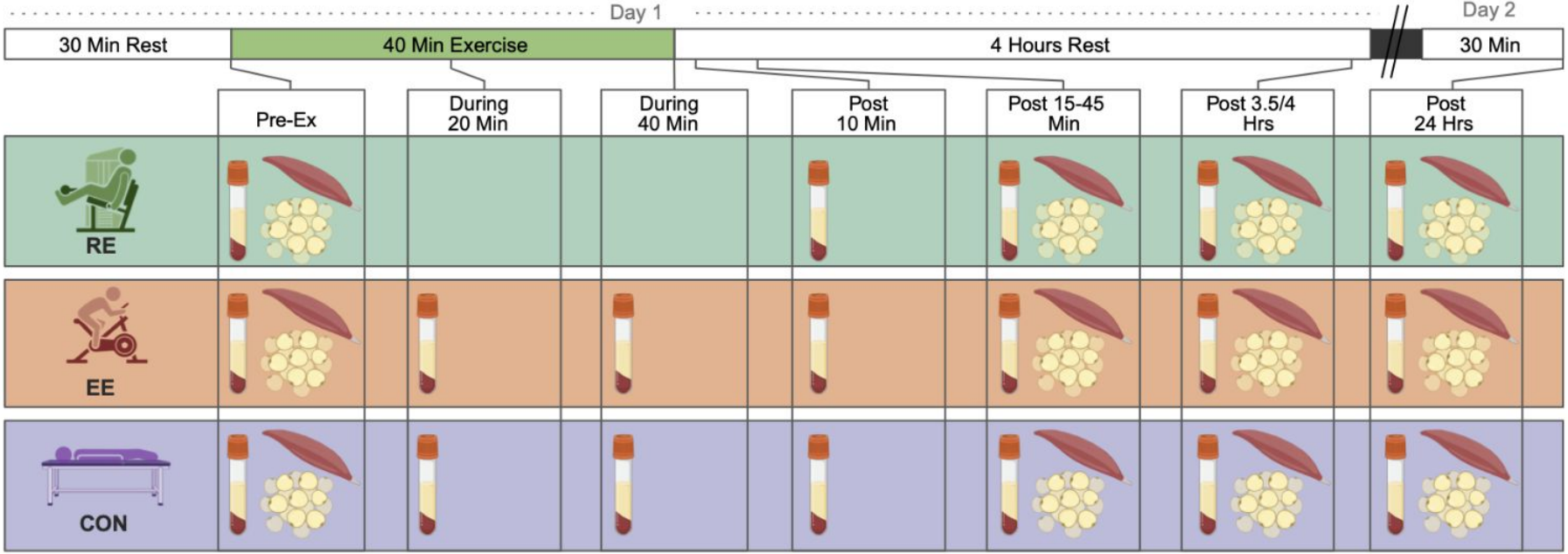
Pill or Pushup

Madza Farias-Virgens, Paul Schwerd-Kleine, Jenn Kerr, Rhea Rodrigues, Joanna Slota, Vylyny Chat

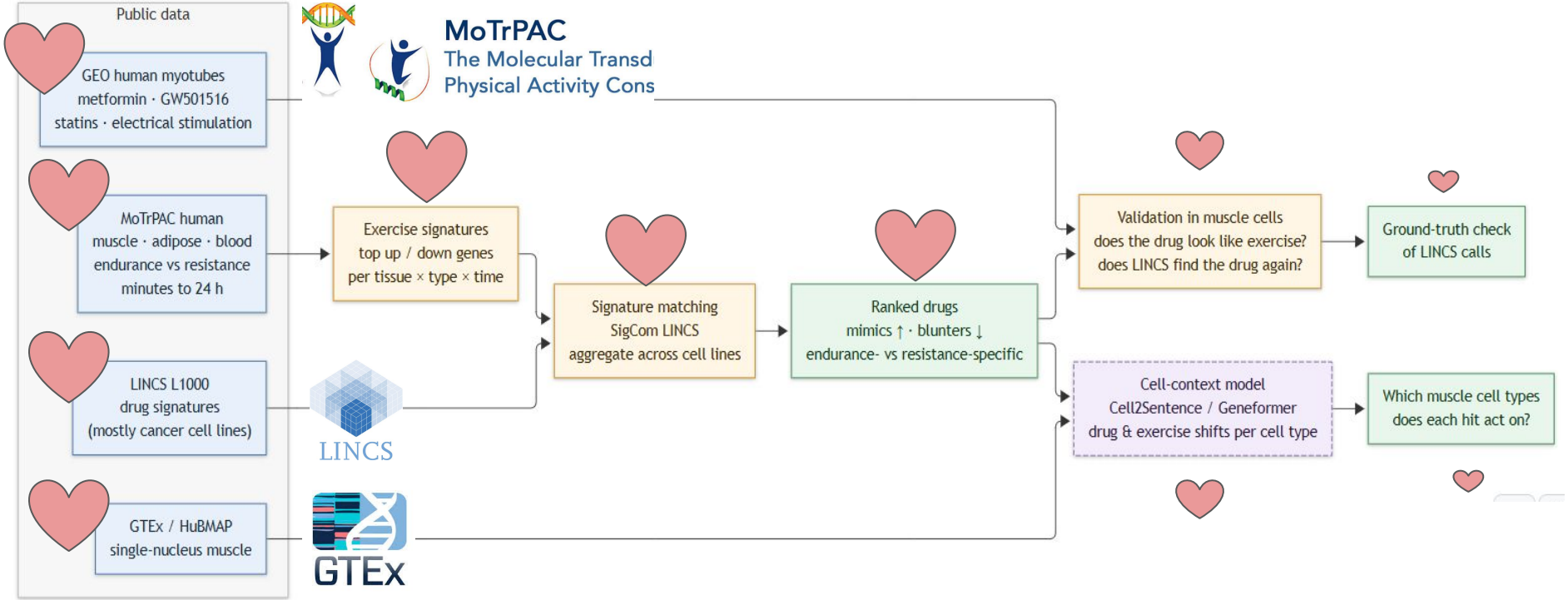

Can we find drugs that mimic/oppose exercise?

MoTrPAC

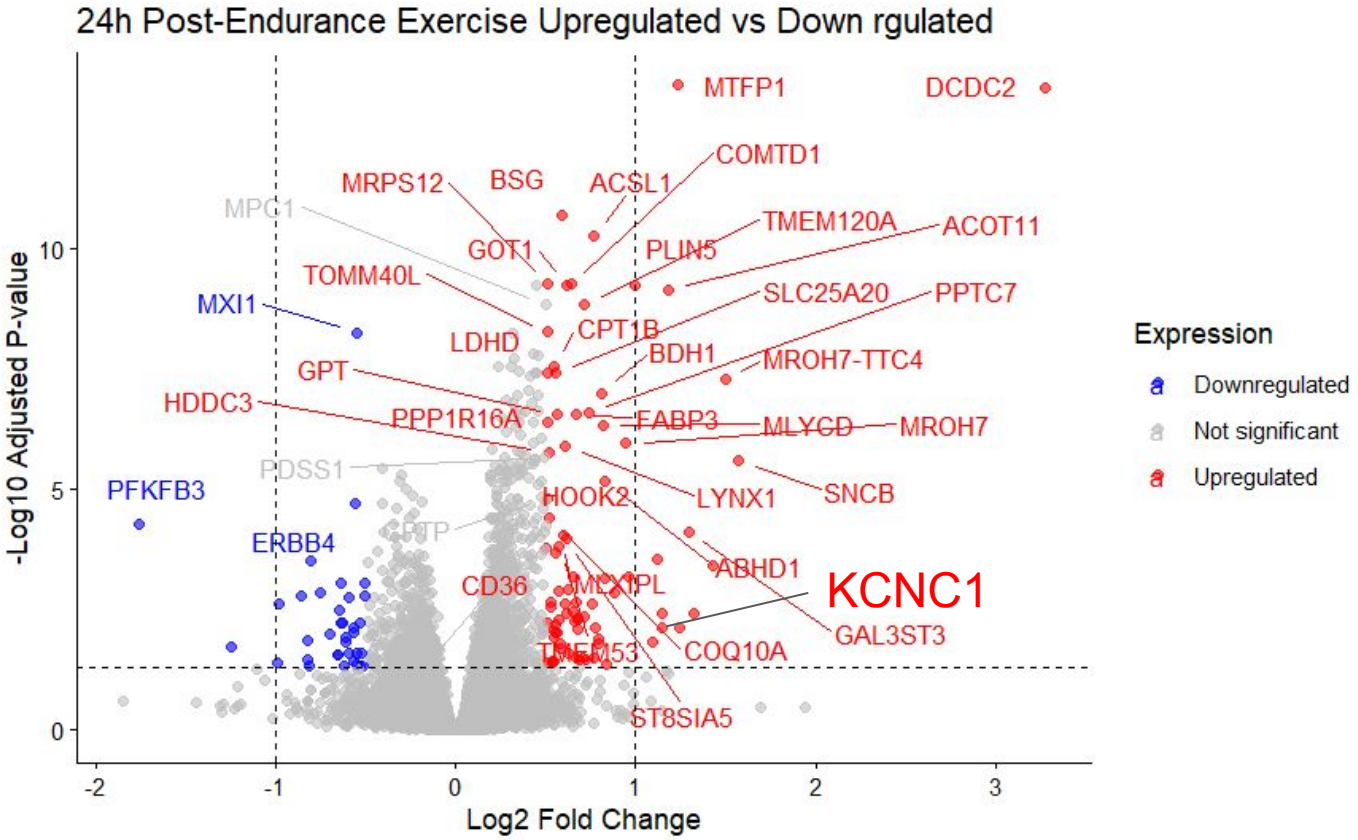
The Molecular Transd
Physical Activity Cons

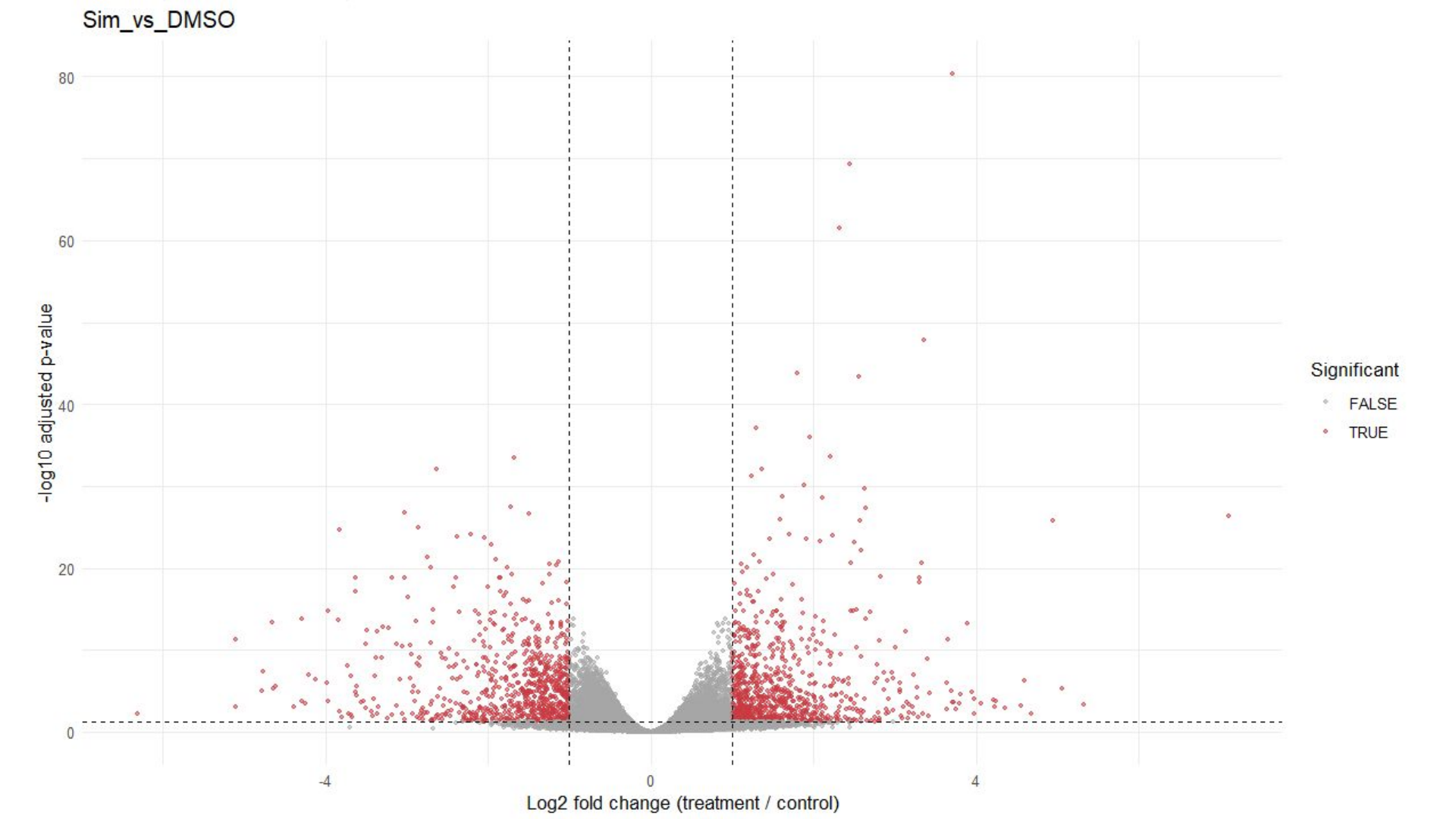


Workflow


Public data
GEO human myotubes
metformin · GW501516
statins · electrical stimulation
MoTrPAC human
muscle · adipose · blood
endurance vs resistance
minutes to 24 h
LINCS L1000
drug signatures
(mostly cancer cell lines)
GTEx / HuBMAP
single-nucleus muscle
MoTrPAC
The Molecular Transd
Physical Activity Cons
Exercise signatures
top up / down genes
per tissue × type × time
LINCS
GTEx
Signature matching
SigCom LINCS
aggregate across cell lines
Ranked drugs
mimics ↑ · blunters ↓
endurance- vs resistance-specific
Validation in muscle cells
does the drug look like exercise?
does LINCS find the drug again?
Ground-truth check
of LINCS calls
Cell-context model
Cell2Sentence / Geneformer
drug & exercise shifts per cell type
Which muscle cell types
does each hit act on?

Genes upregulated and downregulated in MoTRPac dataset




Sim_vs_DMSO
80
60
40
20
0
-log10 adjusted p-value
-4
0
4
Log2 fold change (treatment / control)
Significant
FALSE
TRUE

Sim_vs_DMSO versus exercise



Test	Estimate	P_value	N
Spearman correlation (two-sided)	-0.07513	1.715e-17	12803
Fisher overlap enrichment (one-sided)	1.49400	0.515	12380
Binomial same-direction agreement (one-sided)	1.00000	0.500	1

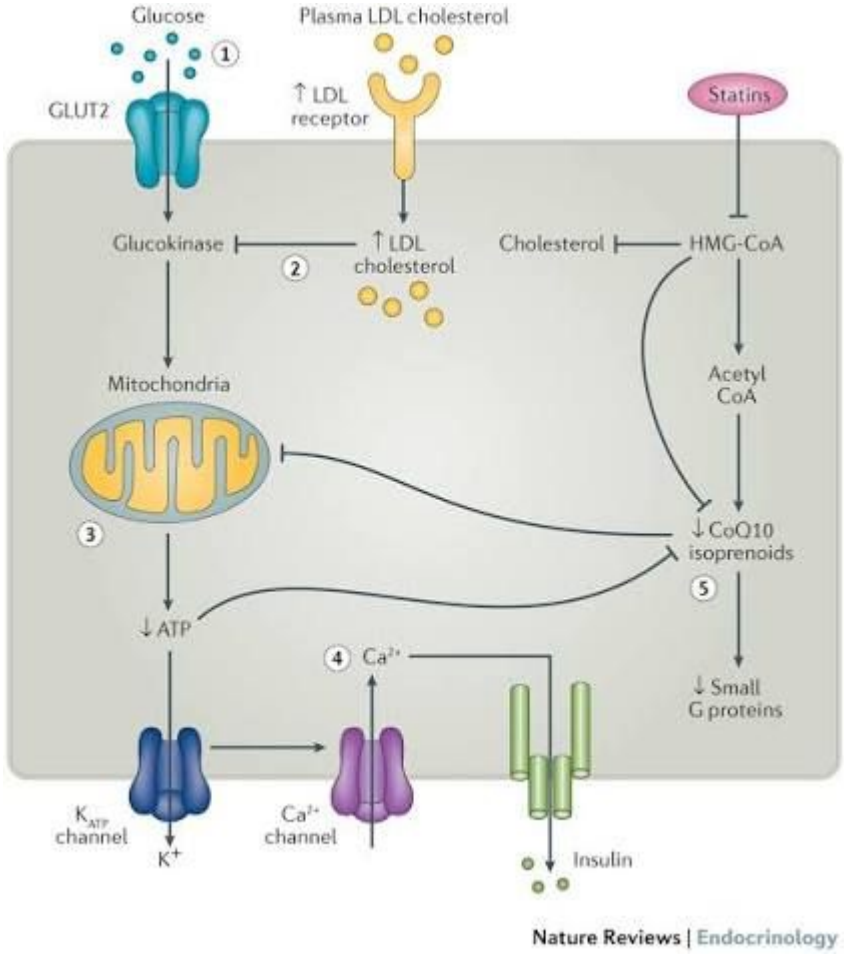
Classification

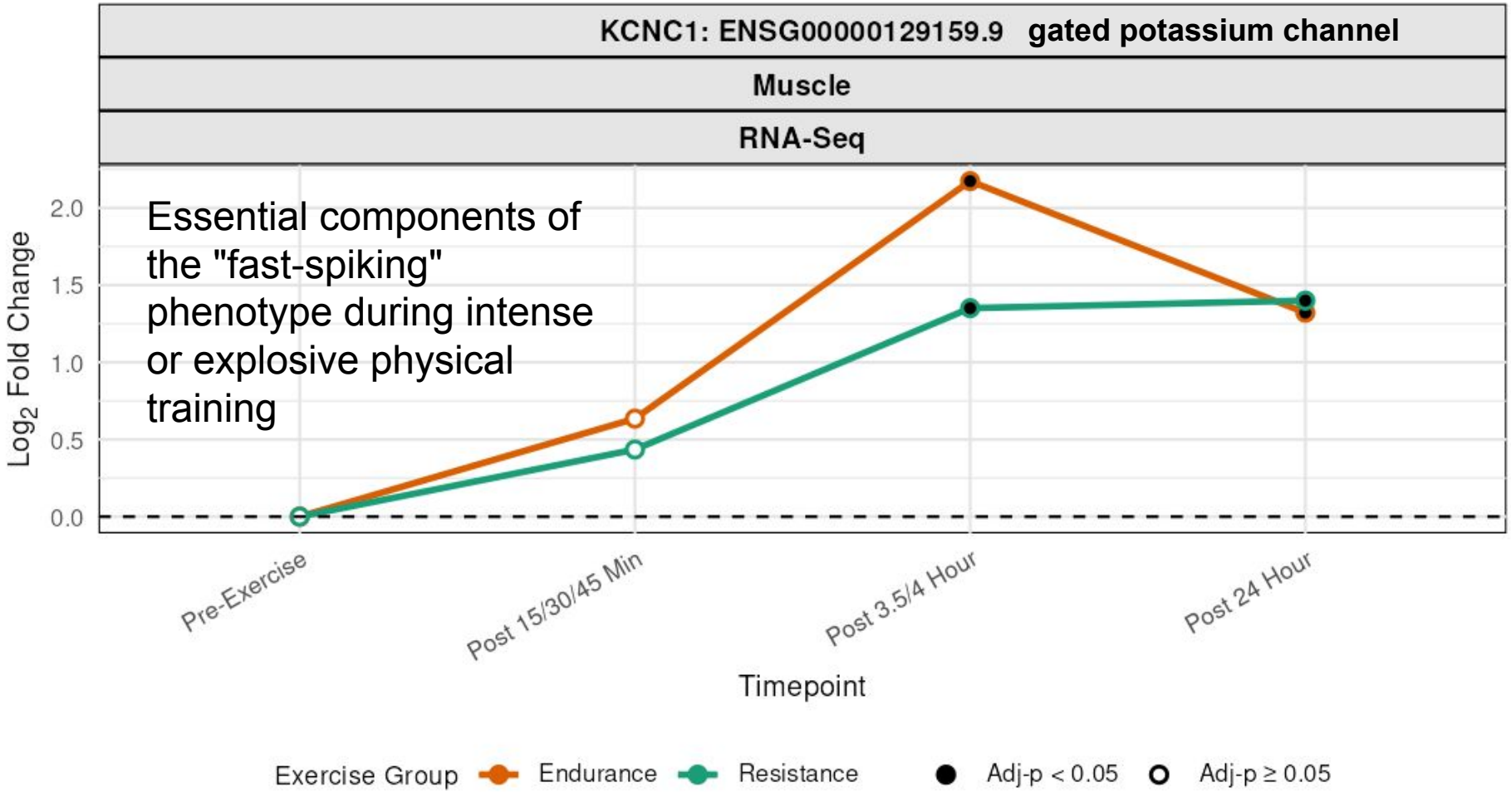
- Both significant: same direction
- Drug only
- Exercise only
- Neither / not evaluable

Simvastatin

- HMG-CoA reductase inhibitor

- It belongs to a class of drugs called HMG-CoA reductase inhibitors. It blocks an enzyme in your liver that is responsible for producing cholesterol




KCNC1: ENSG00000129159.9 gated potassium channel
Muscle
RNA-Seq
Essential components of the "fast-spiking" phenotype during intense or explosive physical training
Log2 Fold Change
2.0
1.5
1.0
0.5
0.0
Pre-Exercise
Post 15/30/45 Min
Post 3.5/4 Hour
Post 24 Hour
Timepoint
Exercise Group
Endurance
Resistance
Adj-p < 0.05
Adj-p ≥ 0.05

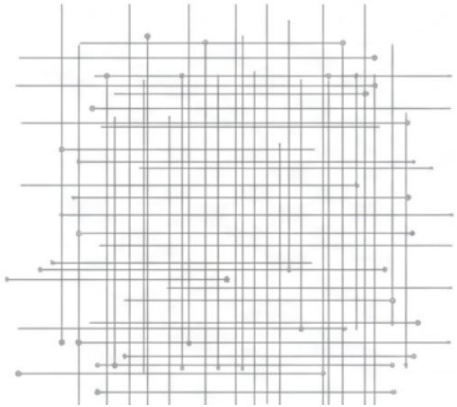
Signature search using DEG from the MoTrPAC (against the LINCS database)

Metric	MoTrPAC (Query)	LINCS L1000 (Target)
Sample Source	Human skeletal muscle (*in-vivo*)	Cultured cancer cell lines (*in-vitro*)
Experimental Scope	24 h post-endurance exercise vs. baseline	136,460 signatures (19,554 compounds)
Gene Panel	Top **100 up / 100 down** DEGs	**10,174 genes** (978 landmark + 9,196 inferred)
Statistical Controls	Adjusted for age, sex, BMI, site, batch & RNA quality	Variable doses & exposure times (6–24 h)

Scoring Drug Compounds Against MoTrPAC Signatures

1. The Database

136,460 drug signatures.
One exemplar per drug × dose × time × cell line.



2. Signature Mapping



Exercise UP genes (es_up)

Exercise DOWN genes

3. WTCS Calculation

```
WTCS = (es_up − es_down) / 2
```

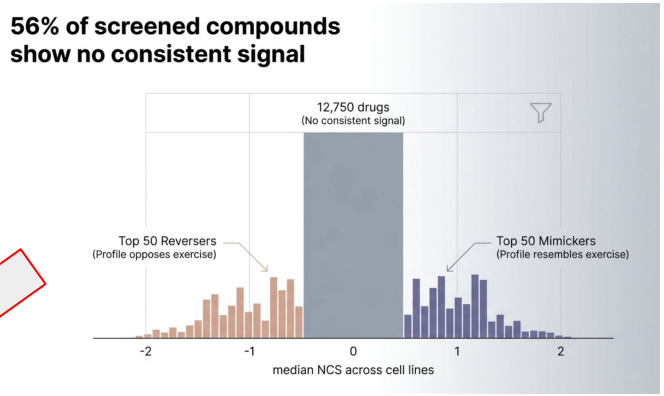
Requires opposing movement. If sets move the same direction, score = 0.

4. Final Output

NCS rescales WTCS within each cell line to calculate the median across cell lines



Candidate exercise mimickers and opposers



Drugs opposing exercise effect

Drugs mimicking exercise effect

The Reversers

Direct Pathway Mirroring

Composition: Highly annotated, targeted compounds.

Annotated Hits: Coherent pathway blockers directly targeting mTOR, PI3K, and IGF-1 networks.

Interpretation: These drugs specifically block the precise growth and recovery pathways activated by endurance exercise.

Hypothetically:

- **Give a Mimicker**: To **simulate** exercise benefits for someone who cannot exercise
- **Identify a Reverser**: To understand what **blocks** exercise benefits or stops muscle recovery pathways (may counteract exercise benefit)

The Mimickers

Growth-Arrest Component

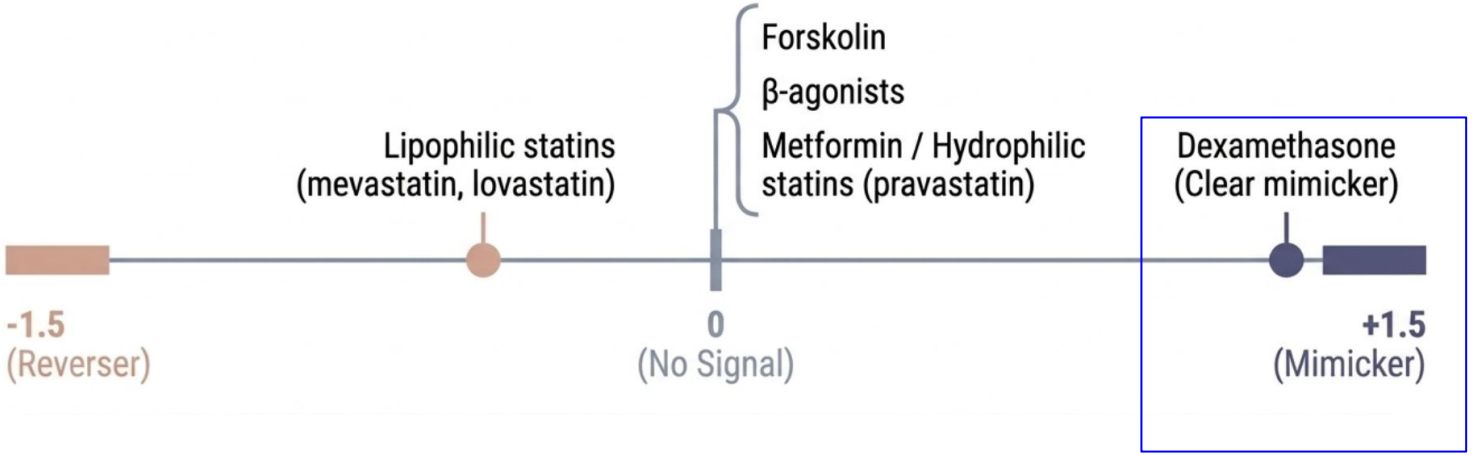
Composition: 43 of the top 50 are unannotated research compounds.

Annotated Hits: Broad anti-proliferative drugs including CDK and DHODH inhibitors.

Interpretation: High scores are largely a mechanistic artifact of shutting down cellular growth and short-lived transcripts.

Can our pipeline recapitulate positive controls (known exercise mimickers)?

Most don't replicate, except for Dexa



Data Caveats:

- LINCS relies mostly on cancer cell lines, which lack normal skeletal muscle metabolic pathways
- Furthermore, acute exercise signals like cAMP and CREB have already returned to baseline by the 24-hour post-workout mark

Can drug signatures shift muscle-cell context in an exercise-like direction?

A cell-type-aware, in-silico pilot using real reference nuclei and public expression signatures

PUBLIC DATA → PAIRED SIMULATED SHIFTS → DIRECTION + CELL-TYPE COMPARISON

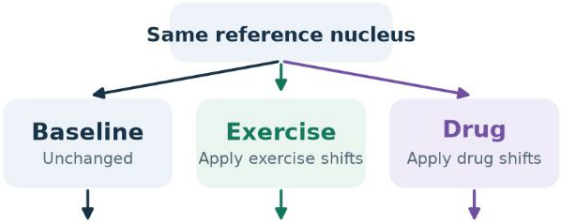
1 Use real biological inputs

GTEx skeletal-muscle nuclei
66 nuclei · 22 annotated cell types
3 nuclei sampled per type

Exercise gene signature
Endurance vs control · 24 h · 16 genes

5 GEO drug signatures
Metformin (2 doses), simvastatin, rosuvastatin and GW501516 / Cardarine

2 Simulate paired shifts

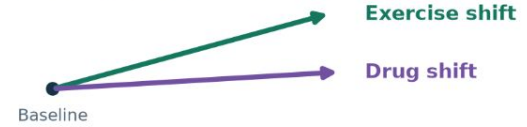


Re-rank the top 300 expressed genes
Gene-rank sentences → C2S embeddings

Bounded heuristic shifts from signed gene scores; not measured post-treatment cells.

3 Compare shift directions

Subtract the same baseline embedding; compare the two resulting shift vectors.

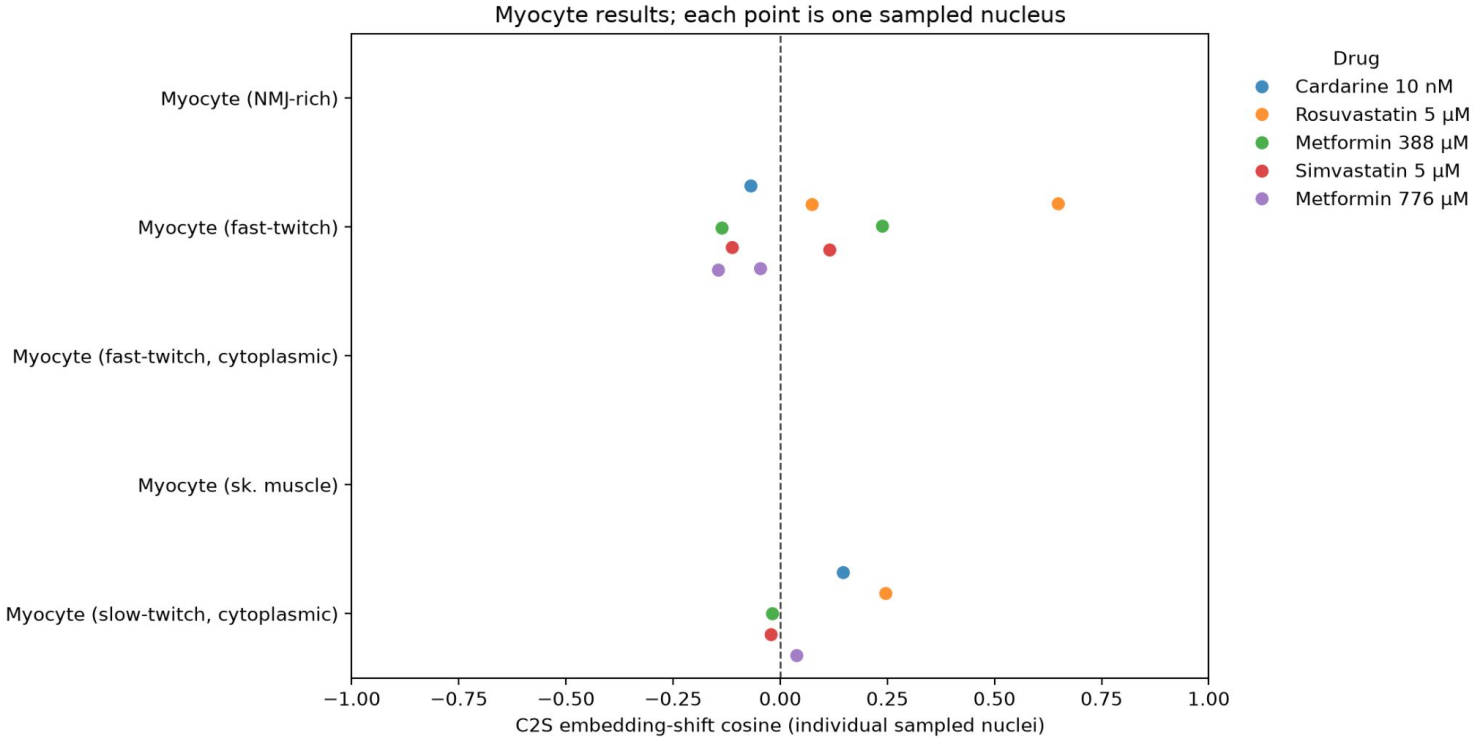


Conceptual geometry, not plotted results

Cosine similarity: Δ_{drug} vs $\Delta_{exercise}$
Average valid paired scores within each cell type

Independent gene-level comparison
Expression-weighted Pearson correlation; requires ≥3 shared, expressed signature genes.

Pilot (16 gene signature, 24h post exercise, not relative to control)



Summary

MoTRPac we identified 117 (82 upregulated , 35 downregulated).

Simvastatin matched with the exercise.

Two main pathways were identified with the initial linc analysis .

Cell2Sentence *in silico* pipeline runs end-to-end. Did not separate signal from noise yet (flawed input signature).